

Middleware Lab: JWT Security Structure Verification

Duration : 30 Min.

Scenario

A security auditor requests verification of incoming JSON Web Tokens (JWT) to ensure client payloads are authenticated.

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Analyze JWT token segments, decode base64 header structures, and verify key claim fields.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Submit Final"** button. This task is evaluated manually by the grading team.

Evaluation Rubric:

1. Identifying Header, Payload, and Signature JWT segments: 3 marks
2. Decoding base64 payloads to verify key claims (sub, exp, aud): 3 marks
3. Describing signature validation processes to verify token integrity: 4 marks

Total Score: 10 Marks

Important Notes

- This is a manually evaluated question.
- Ensure all logs and configurations are fully saved in your workspace folder before submitting.